Sorathorn Thongpitukthavorn

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EDUCATION

Cornell University (Cornell Tech) **Aug 2026 - May 2027**
Master of Engineering, Computer Science *New York, NY*
- **Coursework:** Applied Machine Learning, Optimization Methods, Networks & Markets, Product Studio

Boston University **Sep 2022 - May 2026**
Bachelor of Arts, Computer Science *Boston, MA*

WORK EXPERIENCE

Asia Inter Rice Co., Ltd. **May 2026 - Aug 2026**
Software & Automation Engineer (Contract / Freelance) *Bangkok, Thailand*
- Built and deployed a full-stack property management platform (React, FastAPI, SQLite) on GCP VMs for 60+ residential units, reducing late payments by 80% through automated LINE tenant webhooks. Engineered an OCR pipeline using Google Cloud Vision to extract utility-meter readings and automate billing reminders, saving 20 hours per week of manual billing entry.
- Developed a desktop automation and data-conversion platform (Electron, Flask, Python, Pandas) converting purchase orders across 11 major Thai retailers from heterogeneous formats (PDF, Excel, HTML, XML) into standardized ERP-ready records. Implemented robust data parsing and automated labor costing algorithms, cutting weekly order processing time by >90%.
- Built a financial reconciliation desktop application with four automated workflows and a multi-bank statement engine that parses PDF records from major banks. Implemented zero-lock shared SQLite concurrency to match remittances to outstanding invoices and track overdue balances, streamlining reconciliation and reducing manual errors.

Datapro Computer Systems **May 2025 - Aug 2025**
Software Developer Intern *Bangkok, Thailand*
- Built a cross-platform passport-processing mobile application with React Native (TypeScript), Spring Boot, and PostgreSQL, integrating pdfplumber OCR to automatically scan, extract, and validate travel documents. Eliminated manual data entry, sped up document handling, and deployed the containerized backend on client servers.
- Architected containerized CI/CD pipelines via Docker and GitLab, reducing deployment cycles by 40% while ensuring strict environment parity across staging and production.
- Designed automated integration and regression testing suites with Selenium and Jira, proactively resolving 30+ bugs and edge cases before user acceptance testing to ensure system reliability.

RELEVANT PROJECTS

Skail (github)
- Architected an open-source, local-first multi-agent coding harness built on LangGraph and DeepAgents, orchestrating a lead agent and up to 3 concurrent specialized subagents (Explorer, Implementer, Tester) via typed, dependency-aware DAG task scheduling.
- Engineered isolated Git worktrees for concurrent file-writing agents with serialized change-set integration, deterministic loop stagnation detection, and automated test/build verification gates to validate task outcomes.
- Implemented task-bound capability-floor LLM routing with upfront transactional cost reservations, paired with a high-throughput SQLite persistence journal supporting session recovery and human-in-the-loop interrupts

Project SCYLLA (github)
- Developed an asynchronous, dual-engine quantitative trading and options analytics platform pairing a multithreaded C++ Crow REST microservice with Python market-data pipelines and an embedded native LightGBM C-API inference engine for single and batch predictions.
- Engineered real-time anomaly detection flagging institutional block trades and directional market regimes, backed by a jittered retry pipeline with circuit-breaker fault tolerance
- Developed 5 LightGBM quantile models and validated trading strategies through chronological out-of-sample walk-forward backtesting, incorporating options flow, volatility, trend indicators, synthetic Black-Scholes ensembles, and dynamic Kelly criterion position sizing.

Cloud Infrastructure & Data Pipelines (github)
- Built Docker and GKE (Kubernetes) microservices alongside Apache Beam/Dataflow pipelines processing tens of millions of records, integrating Pub/Sub alerts and least-privilege IAM

TECHNICAL SKILLS

- **AI / Agents & ML**: LangGraph, DeepAgents, Multi-Agent Orchestration, Model Routing, LiteLLM, PyTorch, Scikit-learn, OpenAI API, Prompt Engineering,
- **Languages**: Python, C++, TypeScript, JavaScript, Go, Java, C, Bash, SQL
- **Backend & Systems**: FastAPI, Flask, Spring Boot, REST APIs, WebSockets, Electron, Docker, Linux, GKE, AWS, GCP, Kubernetes
- **Databases & Tools**: PostgreSQL, SQLite, Git, GitLab CI/CD, React Native, Vite, Node.js, Selenium, Pandas, openpyxl, pdfplumber
- **Certifications**: CompTIA Security+, CompTIA Network+, CompTIA A+, CompTIA IT Fundamentals (ITF+)